CSE470

MD IBRAHIM ALIF

ID: 22101221

Sec: 17

- Create A Scenario that requires more than 1 design pattern to Address the scenario then solve it using A convenient programming language (Python / Java). Then write it down using pen, paper

Foodpanda has created a special app only for Bracu. They are using functionalities from the original app. However they are not able to provide everything with original codebase but they can't modify those codes either else main app will fail. They also wanted to send a ping to user when a shop user following is open.

Answer: Need adapter + observer for this.

```
class FoodPanda:

    def importantMethods(self):
        veryImportantwork()


class BracPanda(foodPanda):
    subscribers = []
    def importantMethods(self):
        veryImportantbutBracu()
```

```
    @staticmethod
    def subscribe(user):
        subscribers.append(user)

    def ping(self):
        for user in subscribers:
            Subscriber.getSub(user).getUpdate()


class Subscriber:
    @staticmethod
    def getSub(id):
        return Subscriber.filter(id)

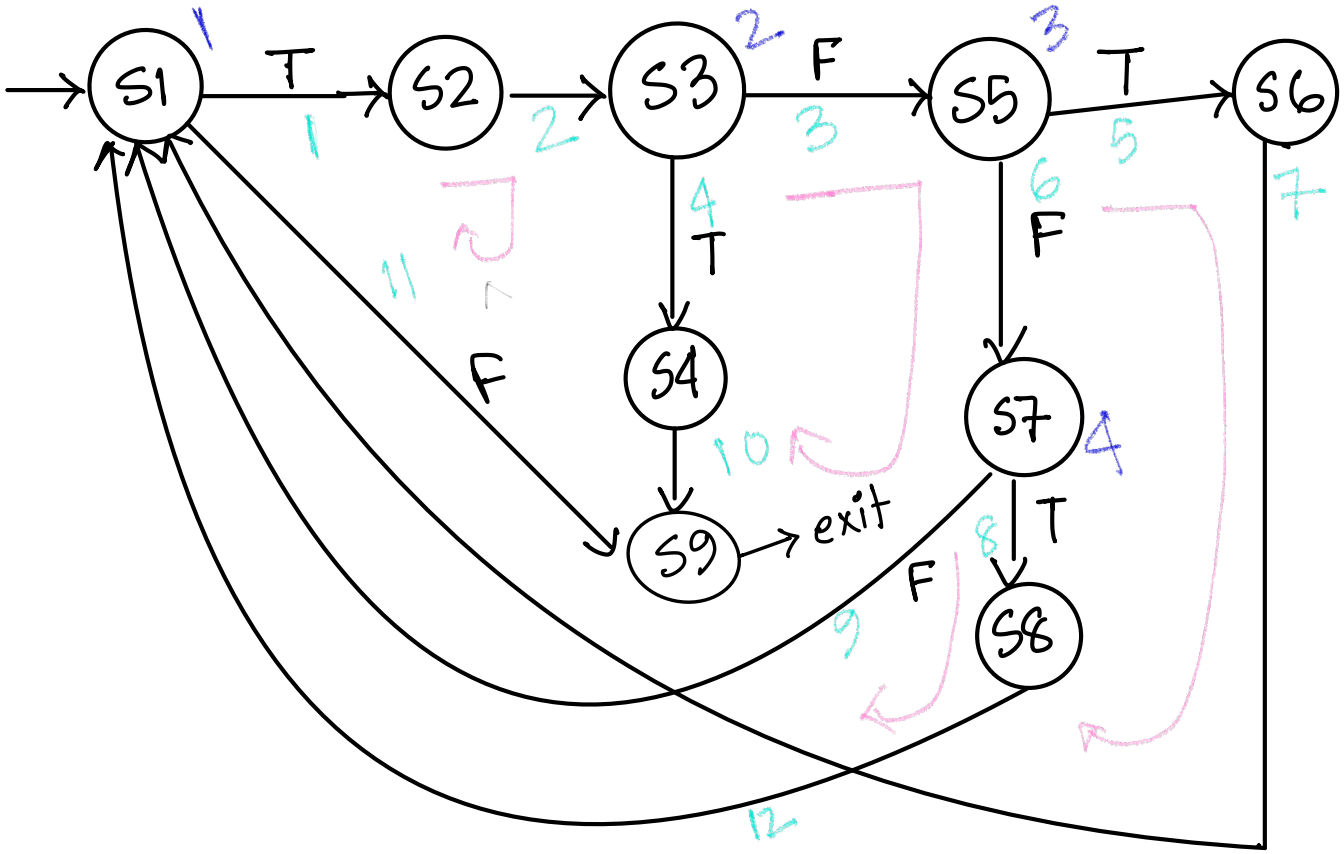
    def getSub(self):
        self.pushNoti()

    def subscribe(self):
        BracPanda.subscribe(user.id)
```

- Create a Control Flow Graph (CFG) Code that carries a scenario where you will have a nested loop as given code like the given practice sheet. Then draw a graph, then calculate Cyclomatic Metric Complexity using All 3 formulas, then identify the independent paths and lastly test those paths using A specific input set.

The following program works as a registration feature for a special club. It takes the age of the members registering for the next event. If the input number is -1, it stops taking input and starts the next process. Else it keeps taking inputs as long as user inputs. However, if one persons age is less than 18, the program prints 'Not eligible'.

```
def register(cap):
    for i in cap:                          S1
        reg = []                           S2
        age = int(input('Enter Age:'))     S2
        if (age==-1):                      S3
            break                          S4
        elif (age<18):                     S5
            print('ineligible')            S6
        elif (age >=18):                   S7
            reg.append(age)                S8
    return reg                             S9
```



$M = R + 1 = 4 + 1 = 5$

$M = P + 1 = 4 + 1 = 5$

$M = E - N + 2P = 12 - 9 + 2 \cdot 1 = 5$

Independent Paths:

1. $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 9$
2. $1 \rightarrow 2 \rightarrow 3 \rightarrow 5 \rightarrow 6 \rightarrow 1 \rightarrow 9$
3. $1 \rightarrow 2 \rightarrow 3 \rightarrow 7 \rightarrow 1 \rightarrow 9$
4. $1 \rightarrow 2 \rightarrow 3 \rightarrow 7 \rightarrow 8 \rightarrow 1 \rightarrow 9$

Test cases:

age = 5, 15, -1

age = -1

age = 21, -1

for age = 21, -1; the path would be:

1 → 2 → 3 → 7 → 8 → 1 → 9